

“ TO CURE **MELANCHOLY**, CUT A CROSS-SHAPED **HOLE** IN THE... **SKULL**... THE **PATIENT** IS TO BE **HELD IN CHAINS**. ”

Roger Frugardi, Italian surgeon, from *Chirurgia*, late 12th century



Surgeon Roger Frugardi's surgical treatise, *Chirurgia* (Surgery), was one of Europe's earliest books on surgery. This illustration shows a hernia operation in progress.

AROUND 1250, PRIEST AND PHYSICIAN Gilbert the Englishman completed the *Compendium Medicinae* (*Compendium of Medicine*). It became one of the most **widely used medical works** of the Middle Ages and was translated from Latin into German, Hebrew, Catalan, and English. The work had separate volumes devoted to the head, heart, respiratory organs, fevers, and women's diseases. In the books, Gilbert also wrote about the **diagnosis of leprosy** by its numbing effect on skin.

In 1266, English friar and scholar **Roger Bacon** completed his *Opus Majus* (*Greater Work*). Ostensibly a plea for church reform, the work included large sections on experimental observation and natural sciences, intended to

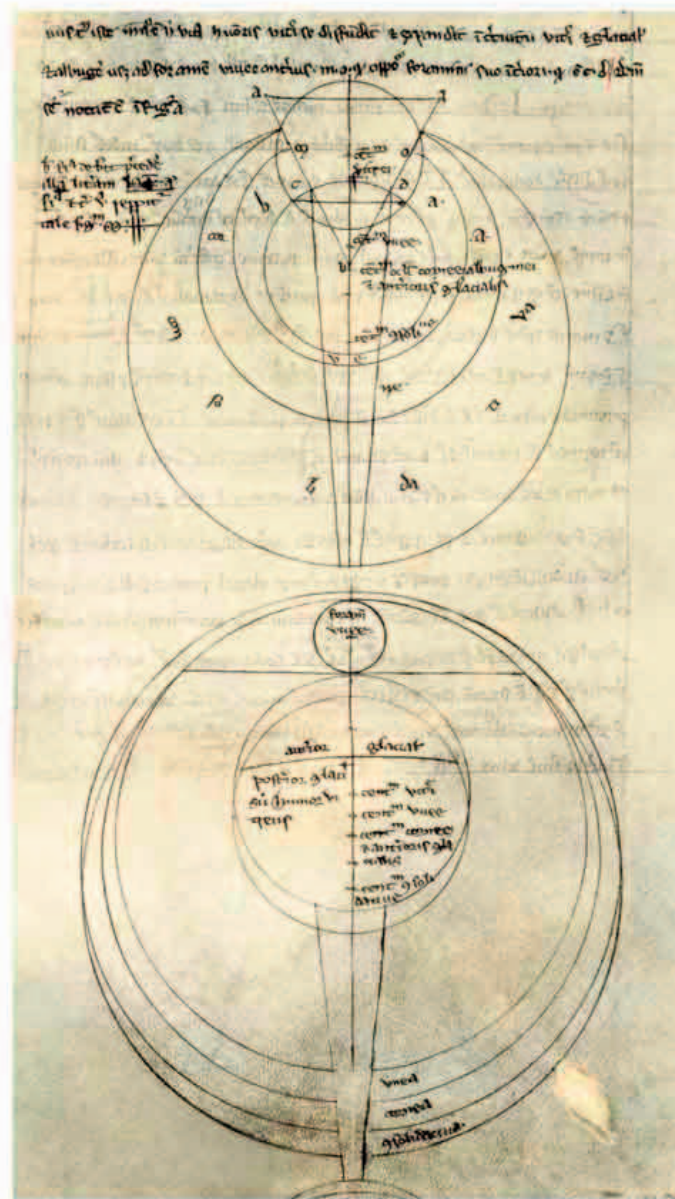
“ EXPERIMENTAL SCIENCE IS THE **QUEEN OF SCIENCES** AND THE **GOAL OF ALL SPECULATION**. ”

Roger Bacon, in *Opus Tertium*, 1267

convince the Church of the **virtues of new learning**. It also contained the first description of **gunpowder** in western Europe, and ideas for flying machines and steamships. The section on **optics** was particularly important. In it, Bacon agreed with Arab scholar Alhazen's view that vision is made possible when rays emanating from the object viewed enter into the eye

(see 1000–29). He examined the properties of differently shaped lenses, and described the use and mathematics behind magnifying lenses—although he did not, as commonly supposed, actually invent eye glasses.

Italian surgeons **Hugo** (c.1180–1258) and **Teodorico Borgognoni** (1205–98) came from a family of doctors who practiced in Bologna, a leading center of medicine. By the 1260s, Teodorico was advocating the **cleansing of wounds with wine** and their rapid closing up. This practice was in contrast to most contemporary medical practitioners who went along with Greek physician Galen's insistence that pus be allowed to form in wounds. The Borgognonis also advocated using **dry bandaging for wounds**—discarding the salves and poultices used at the time. They also used an **early form of anesthesia** by holding sponges soaked with narcotics such as opium or hemlock near the noses of patients who were about to undergo surgery.



Opus Majus

This diagram from Roger Bacon's major work shows the structure of the eye, the curvature of its lens, and how light rays striking the lens produce vision.

ROGER BACON (1220–92)

Educated at Oxford University, England, Roger Bacon travelled to Paris, where he lectured on Aristotle. In 1247, he gave up his post to research privately. He joined the Franciscan Order in 1257 in order to continue his studies. He was commissioned by Pope Clement IV to produce a work on church reform, which led to his *Opus Majus* in 1267.



1250 Gilbert the Englishman's *Compendium of Medicine* includes diagnosis of leprosy

c.1260 Teodorico Borgognoni advocates cleansing of wounds with wine and use of anaesthetic narcotics in operations

1260 Albertus Magnus writes that volcanoes are caused by subterranean winds

1267 Roger Bacon describes the structure of the eye and the properties of magnifying lenses